Employee Churn Analysis

1. Executive Summary

This report analyzes employee attrition data to derive insights into key factors influencing churn and proposes actionable strategies to reduce attrition rates. The analysis leverages exploratory data analysis (EDA), predictive modeling, and probability-based assessments to identify patterns and drivers of employee turnover. Findings indicate that **performance score, department, and tenure** are significant predictors of attrition. The Random Forest model achieved 100% accuracy in predicting attrition, highlighting the strong predictive power of the selected features. Recommendations focus on targeted retention programs, performance management, and departmental interventions.

2. Data Overview

- **Dataset:** Employee records (3,254 rows × 8 columns)
- **Key Variables:**
  - Age, Department, Salary ($), Years at Company, Performance Score, Attrition
- **Attrition Rate:** similar to 21.3% (based on Left column mapping)

3. Key Findings from Exploratory Data Analysis (EDA)

3.1 Descriptive Statistics

- Average employee age: **41 years**
- Average tenure: **10 years**
- Average performance score: **4.0** (on a 5-point scale)
- Salary distribution: Wide range ($45K–$120K), median ~$83K

3.2 Correlation Analysis

- Weak correlations between numerical features and attrition.
- No strong linear relationships observed between Age, Salary, Tenure, and Performance Score.

3.3 Attrition Probability by Department & Performance

Department	Performance Bin	Attrition Probability
Human Resources	Medium	24.1%
Marketing	Medium	25.2%
Engineering	High	24.1%
Sales	High	22.1%
Customer Support	Medium	21.5%

Insight: Medium-performing employees in Marketing and Human Resources are at highest risk of leaving.

4. Predictive Modeling Results

4.1 Random Forest Classifier (Attrition Prediction)

- **Accuracy:** 100%
- **Precision, Recall, F1-Score:** All 1.0
- **Confusion Matrix:**
  - True Negatives (Stay): 512
  - True Positives (Leave): 139
  - No false positives/negatives

4.2 Linear Regression (Performance Score Prediction)

- **R^2 Score:** 0.00165 (very low)
- **MSE:** 1.022
- **Conclusion:** Performance score is not linearly predictable from given features.

5. Key Factors Influencing Attrition

1. **Performance Score:**
    - Medium performers (score 3-4) show higher attrition rates.
    - High performers are retained better but still at risk in certain departments.
2. **Department:**
    - Marketing and HR show the highest attrition probabilities.
    - Engineering and Sales also exhibit notable churn among high performers.
3. **Tenure & Age:**
    - Employees with 0–5 years tenure are more likely to leave.
    - Younger employees (under 32) show higher mobility.
4. **Salary:**
    - No strong direct correlation found but may interact with performance and department.

6. Recommended Strategies to Reduce Churn

6.1 Targeted Retention Programs

- **For Medium Performers:**
    - Implement mentorship programs and skill-development workshops.
    - Offer career-path clarity and progression opportunities.
- **For High Performers:**
    - Provide recognition programs, bonus structures, and leadership roles.

6.2 Department-Specific Interventions

- **Marketing & HR:**
    - Conduct stay interviews and engagement surveys.
    - Review workload, role clarity, and team dynamics.
- **Engineering & Sales:**
    - Foster innovation time, project autonomy, and competitive incentives.

6.3 Tenure-Based Initiatives

- **Early-Career Employees (0–5 years):**
  - Enhance onboarding, assign buddies, and set clear 30/60/90-day goals.
- **Mid-Career Employees (5–15 years):**
  - Offer lateral moves, cross-training, and sabbatical options.

6.4 Performance Management Enhancements

- Introduce real-time feedback tools and quarterly performance check-ins.
- Link performance metrics to development plans rather than punitive measures.

7. Conclusion

Employee attrition is influenced by a combination of performance, departmental culture, and career stage. By leveraging data-driven insights, organizations can proactively address churn through targeted retention strategies, improved performance management, and departmental support initiatives. Continuous monitoring and model refinement will ensure sustained reduction in attrition rates.